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3rd Sem / Electrical

Subject : Electrical Measurement & Instrumentation

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 An Ammeter is a : (CO1)

- a) Secondary Instrument
- b) Absolute Instrument
- c) Recording Instrument
- d) Integrating Instrument

Q.2 Induction type instruments are used for (CO2)

- a) A.C Measurements
- b) D.C. Measurements
- c) Resistance Measurements
- d) Temperature Measurements

Q.3 Output of digital multimeter is (CO3)

- a) Mechanical b) Optical
- c) Electrical d) Analog

Q.4 A.C.R.O can be used to measure (CO3)

- a) A.C. Voltage b) D.C Voltage
- c) Frequency d) Any of the above

Q.5 LVDT is _____ type of transducer. (CO4)

- a) Resistive b) Capacitive
- c) Inductive d) Optical

Q.6 Full form of RTD is. (CO5)

- a) Resistance Thermistor Detector
- b) Resistance Temperature Detector
- c) Round Thermistor Detector
- d) Resistance Thermometer Detector

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 What is Thermocouple. (CO5)

Q.8 Define Voltmeter. (CO1)

Q.9 Define transducer. (CO4)

- Q.10 What is C.T. (CO3)
- Q.11 Expand CRT (CO3)
- Q.12 Provide an example of recording instruments.(CO1)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What is the need of maximum demand indicator? (CO2)
- Q.14 In a two wattmeter method explain the relationship between phase angle and wattmeter reading. (CO2)
- Q.15 Differentiate between LVDT and RVDT. (CO5)
- Q.16 Define error and explain various types of errors. (CO1)
- Q.17 Explain essentials of indicating Instruments. (CO1)
- Q.18 What are the various control of CRO? (CO3)
- Q.19 Write four applications of LCR meter. (CO4)
- Q.20 Discuss importance of transducers in electrical measurement. (CO5)
- Q.21 Write note on Piezoelectric Transducer. (CO4)
- Q.22 Explain Instrument Transformer. (CO3)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Draw and explain construction and working principle of 3 ϕ energy meter. (CO2)

Q.24 With the help of neat diagram, explain the working of CRO? (CO3)

Q.25 Write a short note on any two.

a) Photovoltaic cell (CO4)

b) Strain Gauge (CO4)

c) Thermocouple. (CO5)